ELK - Extended Logic Keys

Third-party integration reference: hotkey slots for AtmosphereAutopilot (and its Approach Autopilot add-on), MechJeb2 and NavUtilities

ELK version **1.1.2** for KSP 1.12.5. Source, releases and README: https://github.com/Rjoande/ELK. License MIT. Document date 2026-10-19.

Supported mods and the versions this document was verified against

AtmosphereAutopilot 1.6.1 by Boris-Barboris: https://github.com/Boris-Barboris/AtmosphereAutopilot. The actions marked "fork" exist as native hotkeys only in the GA-hotkeys fork (https://github.com/Rjoande/AtmosphereAutopilot); the state they act on is upstream, so ELK offers them on the official build too.
Approach Autopilot 1.0.1 (formerly **AtmosphereApproach**, 1.0.0) by Rjoande: https://github.com/Rjoande/ApproachAutopilot. An add-on of AtmosphereAutopilot, so its key sits in the AA tab. ELK recognises either name; with both installed the key goes to AtmosphereApproach, since Approach Autopilot disables itself until the old folder is deleted.
MechJeb2 2.15.3 by the MuMech team: https://github.com/MuMech/MechJeb2.
NavInstruments Continued (NavUtilities) 0.8.1.1 by linuxgurugamer: https://github.com/linuxgurugamer/NavInstruments.

All mods are reached by reflection only: nothing bundled, nothing referenced. A mod that is not installed hides its tab and turns its slots into no-ops. Every key calls the member the mod's own window or native hotkey calls. All keys ship empty. "Slot id" is the node name in the cfg file.

AtmosphereAutopilot - 27 slots

Tab **AtmosphereAutopilot**, file *ELK_AtmosphereAutopilot.cfg*. Options: **aa_autoengage** (true): a Cruise Flight or FBW key pressed while that controller is off switches master and controller on first; off = the key is a no-op until the controller is active. **aa_autoshow** (false): open the AA window when the master is switched on. **aa_vs_step** (1), **aa_alt_step** (50), **aa_hdg_step** (1): bug increments. Column "AA native key": the AA hotkey with the same effect; the tab's Import / Export / Restore buttons move bindings between ELK and AA's *Global_settings.txt*.

Manager

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight	AA native key
AA: Master switch	`AA_MASTER_TOGGLE`	`TopModuleManager.Active (setter) + mainMenuGUIUpdate()`	Master on/off, like AA's native master key.	`master_switch_key`
AA: FD (Cruise <-> FBW)	`AA_FD_TOGGLE`	`TopModuleManager.activateAutopilot(CruiseController | StandardFlyByWire)`	From Cruise Flight to Standard FBW; from anything else to Cruise Flight. Switches the master on when needed.	`fd_switch_key (fork)`
AA: Show/hide window	`AA_GUI_TOGGLE`	`GUIWindow.ToggleGUI() on the manager`	Shows or hides AA's main window, like the native Shift+master key.	-
AA: Standard FBW	`AA_SELECT_FBW`	`activateAutopilot(StandardFlyByWire)`	Makes Standard Fly-By-Wire the active controller (master on if needed).	-
AA: Cruise Flight	`AA_SELECT_CRUISE`	`activateAutopilot(CruiseController)`	Makes Cruise Flight the active controller (master on if needed).	-

Cruise Flight (aa_autoengage applies)

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight	AA native key
CF: Level (LVL)	`AA_CF_LEVEL`	`CruiseController.LevelFlightMode = true`	Lateral mode Level: wings level, holds the current heading.	`level_flight_key`
CF: Heading (HDG)	`AA_CF_HEADING`	`CruiseController.CourseHoldMode = true`	Lateral mode Heading: flies the HDG bug. AA itself refuses heading hold beyond 80 degrees of latitude.	`course_hold_key`
CF: Waypoint (NAV)	`AA_CF_WAYPOINT`	`CruiseController.WaypointMode = true`	Lateral mode Waypoint: navigates to the destination set in the Cruise Flight window.	`waypoint_key`
CF: V/S engage/disengage	`AA_CF_VS_TOGGLE`	`vertical_control = true; height_mode = VerticalSpeed`	Vertical control in V/S mode; pressed again in that mode, disengages it.	`vspeed_select_key (fork)`
CF: ALT engage/disengage	`AA_CF_ALT_TOGGLE`	`vertical_control = true; height_mode = Altitude`	Altitude hold; pressed again in that mode, disengages it.	`altitude_select_key (fork)`
CF: Vertical motion on/off	`AA_CF_VERTICAL_TOGGLE`	`CruiseController.vertical_control`	Vertical control on/off without changing its mode.	`vertical_control_key`
CF: ALT <-> V/S-FPA	`AA_CF_SETPOINT_TYPE_TOGGLE`	`CruiseController.height_mode (+ prev_height_change_mode_by_hotkey)`	Altitude hold to the last rate mode used (V/S or FPA) and back, like AA's native key.	`toggle_vertical_setpoint_type_key`
CF: Keys input mode	`AA_CF_KEYS_MODE_TOGGLE`	`CruiseController.use_keys`	Move the bugs with the stock pitch and yaw keys, on/off.	`switch_key_mode`

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight	AA native key
CF: V/S bug up	AA_CF_VS_UP	desired_vertsetpoint.Value += aa_vs_step	Vertical setpoint up one step (m/s in V/S mode, degrees in FPA mode).	vspeed_up_key
CF: V/S bug down	AA_CF_VS_DOWN	desired_vertsetpoint.Value -= aa_vs_step	Lowers the vertical setpoint by one step.	vspeed_down_key
CF: ALT bug up	AA_CF_ALT_UP	desired_altitude.Value += aa_alt_step	Raises the target altitude (ASL) by one step, in metres. Moves the bug only.	altitude_up_key
CF: ALT bug down	AA_CF_ALT_DOWN	desired_altitude.Value -= aa_alt_step	Lowers the target altitude by one step.	altitude_down_key
CF: ALT bug = current	AA_CF_ALT_SYNC	desired_altitude.Value = vessel.altitude; vertical_control = true; height_mode = Altitude	ALT bug = current ASL altitude, altitude hold engaged (never disengaged).	-
CF: HDG bug up	AA_CF_HDG_UP	desired_course.Value += aa_hdg_step (wrapped 0-360)	Turns the target heading right by one step, in degrees. Moves the bug only.	course_up_key
CF: HDG bug down	AA_CF_HDG_DOWN	desired_course.Value -= aa_hdg_step (wrapped 0-360)	Turns the target heading left by one step.	course_down_key
CF: HDG bug = current	AA_CF_HDG_SYNC	desired_course.Value = FlightGlobals.ship_heading; CourseHoldMode = true	Sets the HDG bug to the current compass heading and selects heading hold.	-

Standard Fly-By-Wire (aa_autoengage applies)

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight	AA native key
FBW: Moderation	AA_FBW_MODERATION	StandardFlyByWire.moderation_switch	Toggles angle-of-attack and G-load moderation, like AA's native moderation key.	moderation_keycode
FBW: Rocket mode	AA_FBW_ROCKET	StandardFlyByWire.RocketMode	Toggles FBW rocket mode.	rocket_mode_keycode
FBW: Coordinated turn	AA_FBW_COORD_TURN	StandardFlyByWire.Coord_turn	Toggles coordinated turns.	coord_turn_keycode

Other modules (flags, no auto-engage)

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight	AA native key
AA: Speed control	AA_SPEED_CONTROL	Prograde ThrustController.spd_control_enabled	Automatic speed (throttle) control on/off; effective only while a controller using it is active.	spd_control_toggle_key
AA: Thrust balancing	AA_THRUST_BALANCING	FlightModel.balance_engines	Toggles automatic thrust balancing between engines.	balancing_toggle_key

Approach Autopilot / AtmosphereApproach add-on (no auto-engage)

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight	AA native key
APR: Arm/disarm	AA_APR_TOGGLE	ApproachController.Armed (setter)	Arms or disarms APR on the runway tuned in NavInstruments, like the add-on's own hotkey: it only flips the arm state, without selecting the controller or touching the master, so whatever is flying keeps flying until localizer capture. No runway tuned: the add-on refuses. Controller not created yet (AA master never switched on for this vessel): nothing. Row shown only with the add-on installed.	apr_arm_key

MechJeb2 - 27 slots on two tabs

Tabs **MechJeb2** and **MechJeb2+**, one file *ELK_MechJeb2.cfg*. Every key acts on the active vessel's master MechJeb on each press (no MechJeb aboard: no-op); a module locked in career is never driven. Options: **mj_autoengage** (true): a Translatron speed or Kill H/S key pressed with the Translatron off switches it on in Keep vertical first. **mj_autoshow** (false): also open the window of the module driven. **mj_trans_step** (1 m/s): speed key increment. **Copy from / to SAS**: copy the ten stock SAS bindings onto the matching SmartASS slots, or back (mapping below). The Translatron is a throttle controller with one mode at a time; the speed keys write its Speed box and live setpoint together, like the window's + / 0 / - buttons.

Tab MechJeb2: MechJeb's own part actions (Speed = 0 and Kill H/S driven directly)

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight
SmartASS: Off	`MJ_SASS_OFF`	`MechJebCore.OnDeactivateSmartASSAction (SmartASS target OFF)`	Disengages SmartASS attitude control.
SmartASS: Kill rotation	`MJ_SASS_KILLROT`	`OnKillRotationAction (Target.KILLROT)`	Cancels rotation and holds the current attitude.
SmartASS: Prograde	`MJ_SASS_PROGRADE`	`OnOrbitProgradeAction (Mode.ORBITAL, Target.PROGRADE)`	Points orbital prograde.
SmartASS: Retrograde	`MJ_SASS_RETROGRADE`	`OnOrbitRetrogradeAction (Target.RETROGRADE)`	Points orbital retrograde.
SmartASS: Normal+	`MJ_SASS_NORMAL_PLUS`	`OnOrbitNormalAction (Target.NORMAL_PLUS)`	Points orbit normal (+).
SmartASS: Normal-	`MJ_SASS_NORMAL_MINUS`	`OnOrbitAntinormalAction (Target.NORMAL_MINUS)`	Points orbit anti-normal (-).
SmartASS: Radial+	`MJ_SASS_RADIAL_PLUS`	`OnOrbitRadialOutAction (Target.RADIAL_PLUS)`	Points radial out, away from the body (navball radial-out marker).
SmartASS: Radial-	`MJ_SASS_RADIAL_MINUS`	`OnOrbitRadialInAction (Target.RADIAL_MINUS)`	Points radial in, towards the body.
Translatron: Off	`MJ_TRANS_OFF`	`OnTranslatronOffAction (Translatron.SetMode(OFF))`	Switches the Translatron off.
Translatron: Keep vertical	`MJ_TRANS_KEEP_VERT`	`OnTranslatronKeepVertAction (SetMode(KEEP_VERTICAL))`	Holds the vertical speed in the Speed box (0 = hover, negative = descent). Throttle only: attitude is yours unless Kill H/S is on.
Translatron: Speed = 0	`MJ_TRANS_ZERO`	`Translatron.trans_spd.Val = 0; Thrust.TransSpdAct = 0`	The window's "0" button: Speed box and setpoint to 0 in the current mode (hover in Keep vertical). Translatron off: Keep vertical first with mj_autoengage, otherwise no-op.
Translatron: Kill H/S	`MJ_TRANS_KILLH_TOGGLE`	`Thrust.TransKillH (+ Translatron.SetMode(KEEP_VERTICAL))`	Kill horizontal speed: in Keep vertical the Translatron also steers against the horizontal velocity, cancelling drift. Translatron on: toggles the flag. Off: with mj_autoengage engages Keep vertical with the flag on, otherwise toggles the flag only.
Translatron: PANIC	`MJ_PANIC`	`OnPanicAction (Translatron.PanicSwitch())`	Launch abort: throttle off, jettisons everything below the last engine stage and fires it (escape motor), 2 s full-throttle climb, then the landing autopilot lands somewhere. Pressed again: cancels.
Landing: Land somewhere	`MJ_LAND_SOMEWHERE`	`OnLandsomewhereAction (Landing.LandUntargeted)`	Starts the landing autopilot with no target.
Landing: Land at KSC	`MJ_LAND_KSC`	`OnLandTargetAction (LandingGuidance.SetAndLandTargetKSC)`	Targets the KSC pad and starts the landing autopilot on it.
Ascent: Autopilot on/off	`MJ_ASCENT_TOGGLE`	`OnAscentAPToggleAction (Ascent.Users add/remove)`	Engages or disengages the ascent autopilot (Ascent Guidance settings).

Tab MechJeb2+: what the part actions do not cover

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight
SmartASS: Node	`MJ_SASS_NODE`	`SmartASS.target = NODE; Engage()`	Points along the next maneuver node. No planned node: nothing happens.
SmartASS: Target+	`MJ_SASS_TARGET_PLUS`	`SmartASS.mode = TARGET; target = TARGET_PLUS; Engage()`	Points at the selected target. No target: nothing happens.
SmartASS: Target-	`MJ_SASS_TARGET_MINUS`	`SmartASS.mode = TARGET; target = TARGET_MINUS; Engage()`	Points away from the selected target. No target: nothing happens.
Translatron: Keep surface	`MJ_TRANS_KEEP_SURF`	`Translatron.SetMode(KEEP_SURFACE)`	Holds the surface speed in the Speed box by throttle alone, pointing prograde or retrograde; never turns the vessel. Kill H/S has no effect here.
Translatron: Keep orbital	`MJ_TRANS_KEEP_ORBIT`	`Translatron.SetMode(KEEP_ORBITAL)`	Same for orbital speed.

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight
Translatron: Speed +step	MJ_TRANS_SPEED_UP	trans_spd.Val += mj_trans_step; TransSpdAct = trans_spd	The window's "+" button: Speed box and setpoint up one step (m/s). Translatron off: Keep vertical first with mj_autoengage, otherwise no-op.
Translatron: Speed -step	MJ_TRANS_SPEED_DOWN	trans_spd.Val -= mj_trans_step; TransSpdAct = trans_spd	The window's "-" button: one step down. Same rules.
Landing: Land at target	MJ_LAND_TARGET	Landing.LandAtPositionTarget(LandingGuidance)	Lands on the position target set in Landing Guidance. Needs a position target and a vessel not landed.
Landing: Abort	MJ_LAND_ABORT	Landing.StopLanding()	Aborts the landing autopilot, only while it is running.
Node: Execute / abort	MJ_NODE_EXECUTE_TOGGLE	Node.ExecuteOneNode(ManeuverPlanner) / Node.Abort()	Executes the next maneuver node; pressed while running, aborts it. No node or zero delta-v: nothing.
MechJeb: Show/hide menu	MJ_MENU_TOGGLE	MechJebModuleMenu.ShowHideWindow()	Shows or hides MechJeb's main menu window.

Copy from / to SAS mapping

Stock SAS slot	SmartASS slot
SAS_STABILITY	MJ_SASS_KILLROT
SAS_PROGRADE	MJ_SASS_PROGRADE
SAS_RETROGRADE	MJ_SASS_RETROGRADE
SAS_NORMAL	MJ_SASS_NORMAL_PLUS
SAS_ANTINORMAL	MJ_SASS_NORMAL_MINUS

Stock SAS slot	SmartASS slot
SAS_RADIAL_IN	MJ_SASS_RADIAL_MINUS
SAS_RADIAL_OUT	MJ_SASS_RADIAL_PLUS
SAS_TARGET	MJ_SASS_TARGET_PLUS
SAS_ANTITARGET	MJ_SASS_TARGET_MINUS
SAS_MANEUVER	MJ_SASS_NODE

NavUtilities (NavInstruments Continued) - 1 slot

Tab **NavUtilities**, file *ELK_NavUtilities.cfg*, no options.

ELK toolbar label	Slot id (cfg node)	Mod member driven	What the key does in flight
NavUtilities: HSI window	NAV_HSI_TOGGLE	NavUtilLibApp.displayHSI() + toolbar icon sync (SetTrue/SetFalse)	Opens or closes the HSI window, like a left click on the NavUtilities toolbar button (icon kept in sync).